

Workforce by Design? The Effect of Targeted Scholarships on College Major Choice

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July 7, 2026

Abstract

This paper studies whether adding a supplemental scholarship for STEM majors to an existing merit-based aid program increases STEM major choice and completion. I examine South Carolina's Math and Science Scholarship Enhancement, which layers an additional \$2,500 per year onto the state's LIFE and Palmetto Fellows merit scholarships for students in an approved STEM field, using student-level data from the University of South Carolina and out-of-state students as a control group. South Carolina residents were 2.47 percentage points more likely to choose an enhancement-eligible major after the policy took effect, an effect concentrated among male students. I also find a decline in double majoring, but no statistically significant increase in STEM degree completion. The results suggest targeted scholarship funding can shift major choice without generating additional graduates in the targeted fields.

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1 Introduction

The number of jobs in the United States in science, technology, engineering, and math (STEM) occupations is projected to increase by more than 10% in the next decade, more than four times the growth rate of non-STEM occupations (U.S. Bureau of Labor Statistics, 2024). But there are concerns that the United States will not have enough qualified workers to fill these jobs given that nearly 20% of the current STEM workforce and more than 40% of doctoral-level scientists and engineers are foreign-born (National Science Foundation, 2024). Governments have a clear incentive to grow the share of STEM graduates in their jurisdictions, as STEM workers tend to earn higher wages (and thus pay more in taxes) than those in non-STEM occupations and their share of the workforce in an economy is positively associated with overall GDP growth (Podobnik et al., 2020; Smith and Cortes, 2022). In an effort to address growing workforce shortages in STEM and other fields, some states have introduced new scholarships for higher education that target specific academic programs identified as critical to the state’s workforce (Nietzel, 2023).

Increasing the number of STEM graduates can be challenging, primarily due to the difficulty of the coursework for students. According to the Harvard Gazette, only 20% of high school graduates are prepared for STEM courses in college (O’Rourke, 2021). The strongest determinants of the intent to major in STEM are math achievement and exposure to math and science courses in high school, although college financial aid offers also play a role (Wang, 2013). Research has shown that common merit-based aid policies that require a minimum GPA to maintain can actually lead to decreases in the number of undergraduates earning degrees with a major in a STEM field (Cornwell et al., 2006; Sjoquist and Winters, 2015). Earning a STEM degree is relatively difficult, and some students prefer to avoid the risk of falling below a GPA threshold and losing their scholarships by simply choosing easier majors. However, a randomized experiment found that offering additional need-based grant assistance (no GPA required to earn/maintain) to low-income students led to higher rates of majoring in STEM and graduating with a STEM degree (Anderson et al., 2023). Research on the US Department of Education’s National Science and Mathematics Access to Retain Talent (SMART) grant program, which provided funding to low-income students and required technical or foreign language majors, has shown that eligible students were 3-10 percentage points more likely to major in the targeted

fields (Denning and Turley, 2017).

In this paper, I study the effect of a supplemental scholarship targeting STEM majors on the probability of a student choosing one of the eligible majors. Specifically, I examine the Math and Science Scholarship Enhancement program in South Carolina, a unique program that functions as an enhancement to pre-existing merit-based awards that require a minimum GPA to maintain. From a policy perspective, it is important to understand whether the additional aid being offered is sufficiently attractive to counteract the potentially negative effect of the GPA requirement. If more students decide to avoid STEM in order to maintain their GPA and protect their initial merit-based aid than the number of students now selecting into STEM to receive the new enhancement, then the net effect of the entire scholarship program on the number of STEM majors would be negative. The effect on STEM graduates would be ambiguous, however, as it depends on the effect on the number of majors as well as the extent to which the enhancement increases the graduation rates of students majoring in STEM fields. Williams (2019) found that the enhancement did increase GPAs and graduation prospects among low-income students, but the author was only able to identify effects on students whose major decisions were made prior to the enhancement program being enacted.

This paper offers evidence on the question of whether introducing a scholarship enhancement that targets specific majors to preexisting merit-based awards that require a minimum GPA to maintain leads to an increase in the number of students pursuing those majors and eventually graduating with degrees. I use student-level enrollment and financial aid data from the University of South Carolina (USC) to estimate the effect of the 2007-08 scholarship enhancement program on both STEM majors and graduation rates by treating out-of-state students (who are not eligible for the scholarship) as a control group. I find that SC resident students were 2.47 percentage points more likely to choose a major eligible for enhancements after the policy, largely driven by male students. There is no evidence that these students are self-sorting into STEM majors with relatively higher pre-enhancement average GPAs. I also find a slight decrease in the probability of SC resident students choosing to double major, perhaps due to time constraints or fear of missing the 3.0 GPA requirement necessary to maintain funding. However, I ultimately detected no statistically significant evidence of an increase in students graduating with STEM degrees, suggesting that additional scholarship funding for targeted fields is not sufficient for generating new graduates.

The remainder of the paper is organized as follows: Section 2 provides information on the scholarship program in South Carolina. Sections 3 and 4 outline the data and the empirical strategy, respectively. Section 5 discusses the main results. Section 6 concludes and offers avenues for future research.

2 Lottery Scholarships in South Carolina

Since its inception in 2002, the South Carolina Education Lottery has awarded more than 2.7 million individual scholarships and transferred more than \$8 billion to the General Assembly for supporting education in the state (Litchfield, 2024). A significant share of funding supports students through two merit-based scholarships, the Legislative Incentive for Future Excellence (LIFE) scholarship and the Palmetto Fellows scholarship. Potential recipients of either must first be South Carolina residents enrolled in eligible SC institutions. There is no application process, but students must meet a combination of criteria involving high school performance and standardized test scores in order to qualify for the aid.¹ The Palmetto Fellows scholarship is both the most difficult to achieve and the most valuable (\$6,700 in year 1, \$7,500 in years 2-4). Though a LIFE scholarship is easier to earn and provides less value (\$5,000 in years 1-4), both scholarships require a minimum college GPA of 3.0 and 30 earned credit hours per year in order to maintain funding. Both scholarships are also limited to a maximum of eight semesters.

In an attempt to boost the state’s STEM workforce, the General Assembly passed legislation in 2007 establishing scholarship enhancements for LIFE or Palmetto Fellows recipients majoring in a math or science program approved by the South Carolina Commission on Higher Education. There is no application process or requirement for students to accept or deny the award. Enhancements are automatically given during an eligible student’s sophomore year and provide an additional \$2,500 per year (three year max), valued at roughly 30% of 2007-08 tuition and fees at the flagship campus (South Carolina Commission on Higher Education, 2011). Table 1 below highlights the majors that were initially approved for the scholarship en-

¹LIFE recipients must meet two of the following criteria: 3.0 SC high school GPA, rank in the top 30% of the graduating class, and/or at least an 1100 SAT or 22 ACT score. Palmetto Fellows recipients must meet two of the following criteria: 3.5 GPA, rank in the top 6% of their class in either 10th, 11th, or 12th grade, and/or at least a 1400 SAT or 31 ACT score.

hancement.

Table 1: 2007-08 Approved Programs, University of South Carolina

Biochemistry and Molecular Biology
Biological Sciences
Biomedical Engineering
Cardiovascular Technology
Chemical Engineering
Chemistry
Civil Engineering
Computer Engineering
Computer Information Systems
Computer Science
Electrical Engineering
Environmental Science
Exercise Science
Geological Sciences
Geophysics
Information Science
Integrated Information Technology
Marine Science
Mathematics
Mechanical Engineering
Middle Level Education, Mathematics/Science
Nursing, BSN Completion (RN to BSN)
Nursing, BSN Generic (No RN)
Pharmaceutical Studies
Public Health
Physics
Statistics

Notes: This table lists STEM-related degree programs approved at the USC Columbia campus prior to the 2016–17 academic year. Some program names have been shortened for clarity. Nursing entries reflect different program types.

In the face of a growing teacher shortage both nationally and within the

state, South Carolina recently expanded the program significantly by extending the aid to education majors starting in the 2024-2025 school year (Budds, 2024). While students who are potentially willing to become education majors likely differ in several important ways compared to students who are potentially willing to become STEM majors, the results of this paper will still offer some insights for policymakers about the expected outcomes of this latest expansion and any others being considered for the future.

3 Data

The data for this paper cover all undergraduate students enrolled at the University of South Carolina at the main campus in Columbia between school years 2004-05 and 2013-14. Enrollment information was provided by the University of South Carolina’s Office of Institutional Research, Assessment, and Analytics (OIRAA). Enrollment data is at the student-term level and includes college information (academic period, classification level, semester GPA, cumulative GPA, course hours carried, hours earned, major choice, degree type, and graduation date), demographics (county of residence, gender, race, and birth year), and academic backgrounds (high school attended, SAT/ACT score).² Summary statistics for the sample are shown in Table 1. Relative to the out-of-state group, SC resident students have lower test scores and GPAs, but are more likely to be STEM majors and to graduate from the University of South Carolina. SC residents are also relatively less White, which mostly reflects the in-state population’s larger share of Black students.

4 Methodology

I identify the impact of the 2007-08 scholarship enhancement on major choice and likelihood of graduating with a STEM degree by comparing outcomes for South Carolina residents who are at least sophomores before and after the expansion with nonresidents, who are ineligible for the scholarship program. The basic model being estimated takes the following form:

²County of residence and high school attended are available in the data for SC residents only. Additionally, SAT/ACT scores are missing for approximately 20% of the sample.

$$Y_{it} = \beta_1 SC_i \cdot Post_t + \beta_2 SC_i + \alpha_1 \mathbf{X}_i + \tau_t + \epsilon_{it} \quad (1)$$

where Y_{it} indicates whether the major for student i in term t is eligible for enhancement (or whether the student double-majored); SC_i is a dummy variable equal to 1 if student i is a resident of South Carolina; $Post_t$ is an indicator equal to 1 if student i is observed in the Fall of 2007 or later; $\mathbf{X}_i$ is a vector of personal characteristics (race, gender, birth year, classification level, SAT score, and ACT score); and ϵ_{it} is the error term. β_1 is the coefficient of interest, representing the effect of the enhancement program on college major decisions.

In order for the model to have internal validity and provide plausibly causal estimates, the parallel trends assumption must be satisfied. In this context, it must be the case that there are parallel trends in rates of majoring in approved STEM fields between residents and non-residents prior to the enhancement program and the parallel trends would have continued in the absence of the program. So long as this is true, non-residents can serve as a control group and represent what would have happened among residents had the enhancement never gone into effect.

For the likelihood of graduating with a STEM degree outcome, I estimate the following linear probability model:

$$Grad_i = \delta_1 SC_i \cdot EYears_i + \delta_2 SC_i + \delta_3 EYears_i + \sum_{i=2004}^{2010} \delta_{4i} Cohort_i + \gamma_1 \mathbf{X}_i + u_{it} \quad (2)$$

where $Grad_i$ is an indicator variable equal to 1 if student i graduated with a STEM degree; SC_i indicates a resident of South Carolina; $EYears_i$ allows for cumulative effects of aid by capturing the number of school years student i is observed in after the enhancement program was introduced; $Cohort_i$ is a vector of school years representing the first school year student i was enrolled in college (summation ends at 2010 to allow all cohorts eight semesters to graduate); $\mathbf{X}_i$ is a vector of personal characteristics (race, gender, birth year, classification level, SAT score, and ACT score); and u_{it} is the error term. δ_1 is the coefficient of interest, representing the effect of the enhancement program on the likelihood of graduating with a STEM degree.

Similar to equation (1), causal identification in this case also relies on the parallel trends assumption. Prior to the 2007-08 school year, rates of graduating with a STEM degree must be evolving similarly for both residents

and non-residents of South Carolina. Results from this analysis will be causal only under the assumption that these parallel trends would have continued had the enhancement program never been enacted. In this paper, I use event study analysis and visual inspection of plotted data to argue in defense of the parallel trends assumptions being made.

5 Results

Table 2 provides the results of the estimation of equation (1). Columns (1) and (2) report the estimated effect of the enhancement program on the likelihood of choosing a STEM major without and with controls, respectively. These results suggest that relative to out-of-state students, SC residents were 2.47 percentage points more likely to choose a major eligible for enhancements in response to the program. Column (3) adds an interaction term to explore differential effects by gender. The results suggest that the policy had a larger effect on male students relative to female students. Figure 3 presents event study estimates of the relationship between the enhancement program and the probability of choosing a STEM major for each semester in the sample, indicating a slight increase in impact over time.

Given that I estimate an increase in students choosing STEM broadly in response to the program, it may be reasonable to think certain STEM majors are more likely to see increases than others. In order to protect their base scholarships and new enhancement funding by maintaining a 3.0 GPA, students induced into STEM by the program may try to choose relatively easier STEM majors and avoid the hardest options. I explore this possibility by splitting all eligible STEM majors into difficulty quartiles based on the average GPA of students in the major prior to the enhancement program, which can be seen in Figure 2. Column (1) of Table 3 focuses on STEM majors in the top 25% of pre-enhancement average GPAs while column (2) considers the bottom 25%. The results indicate statistically significant increases in both types and the effect is actually stronger for the STEM majors with the lowest prior GPAs, meaning there is no evidence of students selecting into “easier” STEM majors even if prior GPAs in the major are relatively high.

The enhancement program could also impact the likelihood of students choosing to double major. A student on the margin of adding a second major in a STEM field when their first major is non-STEM would have scholarship money to gain in doing so, making double majoring more likely in response

to the program. On the other hand, a student already double majoring in either one or two STEM fields would not risk any scholarship money by dropping to just one major in STEM and would have an easier time maintaining a 3.0 GPA to ensure access to the scholarship enhancement, meaning double majoring may be less likely. Table 4 considers double majoring as an outcome and explores differential effects by gender. The results suggest that the likelihood of double majoring decreased, particularly among male students.

Finally, while the enhancements did generate additional STEM majors among SC students, perhaps a more policy-relevant outcome is whether the program generated any new STEM *graduates*. Table 5 explores this possibility by limiting the sample to the last observed semester for every student and estimating the program’s effect on the likelihood of graduating with a STEM degree. Whether looking broadly or across gender, there is no statistically significant effect of enhancements on graduating in STEM.

6 Conclusion

This paper offers new evidence on the impact of targeted scholarship enhancements on students’ academic choices. I find that South Carolina’s Math and Science Scholarship Enhancement increased the probability that in-state students majored in approved STEM fields by 2.47 percentage points, largely driven by male students. However, I am unable to detect any graduation effects as a result of the policy. This implies that while financial incentives can induce changes in major choice, it is unclear whether they are sufficient to generate graduates of these major programs. It is possible that the enhancements attracted students who were relatively less-prepared for STEM coursework, leading to a decline in graduation likelihood that offsets the increase in STEM majors.

There is also no evidence of students self-sorting into “easier” STEM majors, though pre-enhancement average GPA by major is an imperfect proxy for major difficulty. It may be the case that STEM majors with the highest pre-enhancement GPAs are not relatively easier but tend to attract the best students, which would explain why students do not appear to concentrate in these fields as a result of the enhancements. Finally, I find a decrease in the likelihood of double majoring. This could reflect risk aversion among students trying to protect their new scholarship funding by simplifying their schedule and maximizing their likelihood of earning at least a 3.0 GPA (re-

quired to maintain eligibility). It could also be a result of time pressure, as the enhancement rules state that funding is only available for up to eight semesters. Many students may have decided that this was not sufficient time to complete two degrees.

As South Carolina and other states across the country continue to expand or experiment with targeted scholarships, it is important for policymakers to consider all of the potential consequences, intended or otherwise. Even though these kinds of policies can be successful in shifting students into the targeted fields while in school, they may not be enough to generate additional graduates of those fields. In addition, special care should be taken when determining scholarship rules and eligibility requirements. In particular, requiring a minimum GPA to maintain funding and capping the number of semesters that can be funded may discourage students from attempting more than one major or taking other academic risks that they otherwise would have taken. Future research should consider targeted scholarship programs paired with other services (mentoring, tutoring, etc.) to explore ways to better support student success in targeted fields.

7 Tables

Table 1: Summary Statistics by South Carolina Residency

	SC Resident	Out-of-State	Difference	P-value
Female	0.54	0.56	0.02	0.000
Year of Birth	1987	1988	0.8	0.000
SAT Total	1100	1183	83	0.000
ACT Composite	23	26	3.2	0.000
Hours Carried	14.1	14.2	0.2	0.000
Hours Earned	12.8	13.4	0.5	0.000
Cumulative GPA	2.994	3.220	0.226	0.000
Enhancement Major	0.319	0.246	-0.073	0.000
Graduated	0.675	0.605	-0.069	0.000
White	0.715	0.773	0.059	0.000
Black	0.159	0.060	-0.099	0.000
Unknown	0.050	0.088	0.037	0.000
Asian	0.032	0.026	-0.007	0.000
Multi	0.014	0.014	0.000	0.556
American Indian	0.003	0.003	-0.000	0.387
Pacific Islander	0.000	0.001	0.000	0.136
Hispanic	0.026	0.036	0.010	0.000
Observations	49,323	30,598		

Notes: Summary statistics calculated at the student level using first observed semester. Sample is restricted to unique students based on ID. The table compares South Carolina resident students and out-of-state students. Standard errors are omitted for brevity.

Table 2: Impact of Enhancements on STEM Choice

	(1)	(2)	(3)
	STEM Major		
Post $\times$ Carolinian	0.0285*** (0.00421)	0.0247*** (0.00417)	0.0331*** (0.00619)
Post $\times$ Carolinian $\times$ Female			-0.0152* (0.00832)
Observations	312,211	312,211	312,211
Demographic Controls	No	Yes	Yes
Semester Fixed Effects	Yes	Yes	Yes

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3: Impact Within STEM: Proxied Difficulty

	(1)	(2)
	Easier STEM	Harder STEM
Post $\times$ Carolinian	0.00434** (0.00215)	0.0132*** (0.00247)
Observations	312,211	312,211
Demographic Controls	Yes	Yes
Semester Fixed Effects	Yes	Yes

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: The Effect of Enhancements on Double Majoring

	(1)	(2)
	Double Major	
Post $\times$ Carolinian	-0.00982*** (0.00141)	-0.0106*** (0.00209)
Post $\times$ Carolinian $\times$ Female		0.00142 (0.00281)
Observations	312,211	312,211
Demographic Controls	Yes	Yes
Semester Fixed Effects	Yes	Yes

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: The Effect of Enhancements on Graduating in STEM

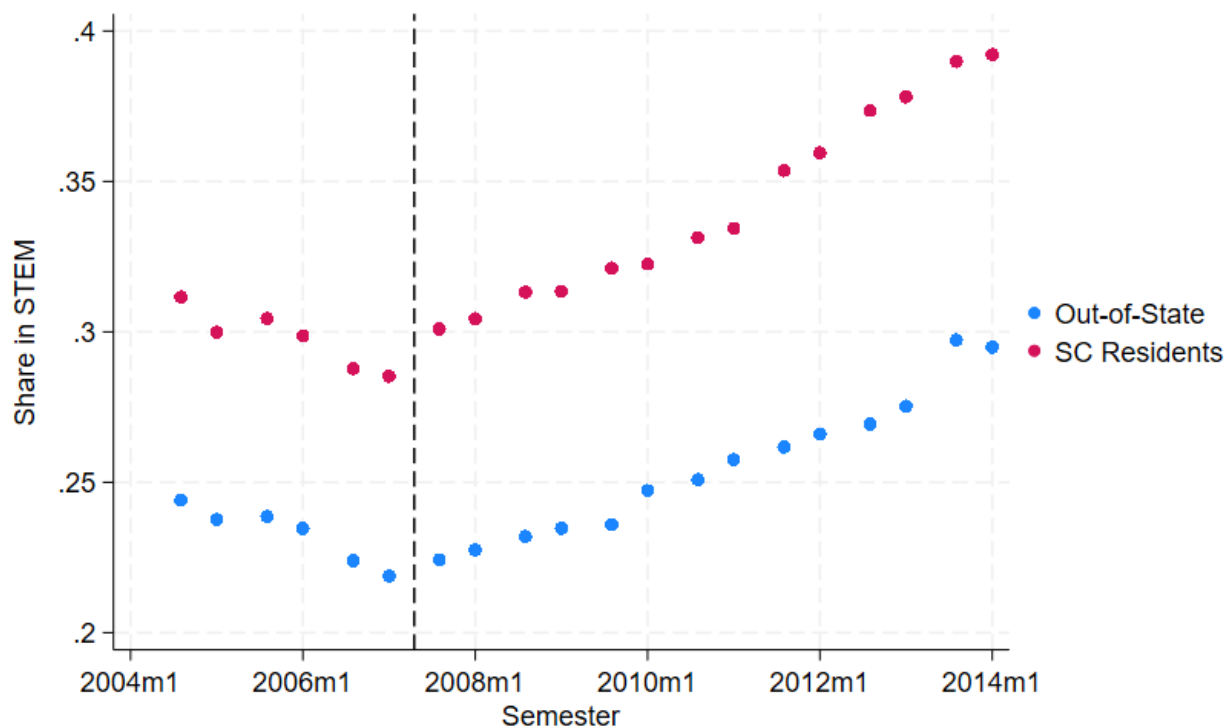
	(1)	(2)
	STEM Graduate	
Post $\times$ Carolinian	0.000669 (0.00171)	-0.00168 (0.00247)
Post $\times$ Carolinian $\times$ Female		0.00403 (0.00331)
Observations	54,425	54,425
Demographic Controls	Yes	Yes
Cohort Fixed Effects	Yes	Yes

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

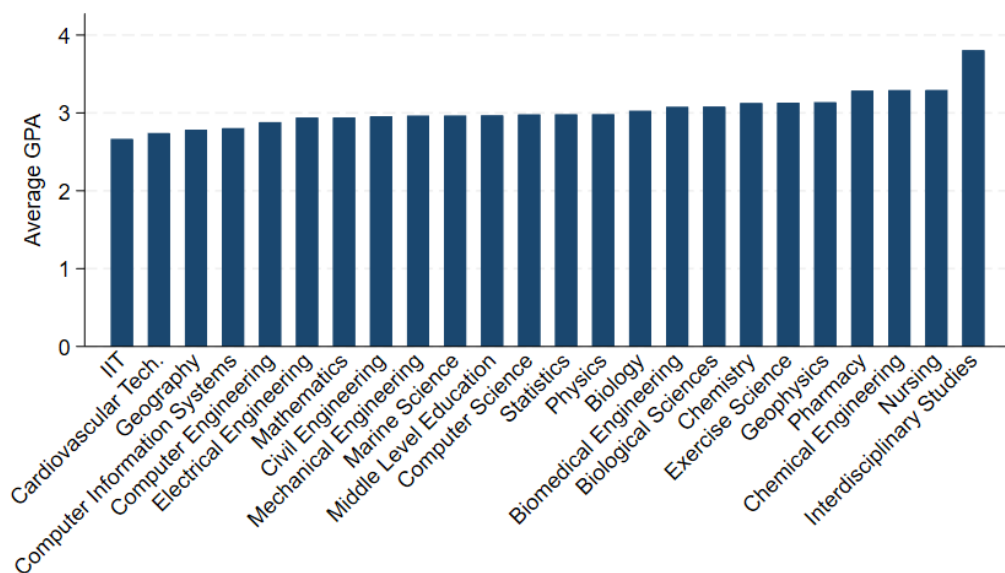
8 Figures

Figure 1: Share of STEM Majors by Residency



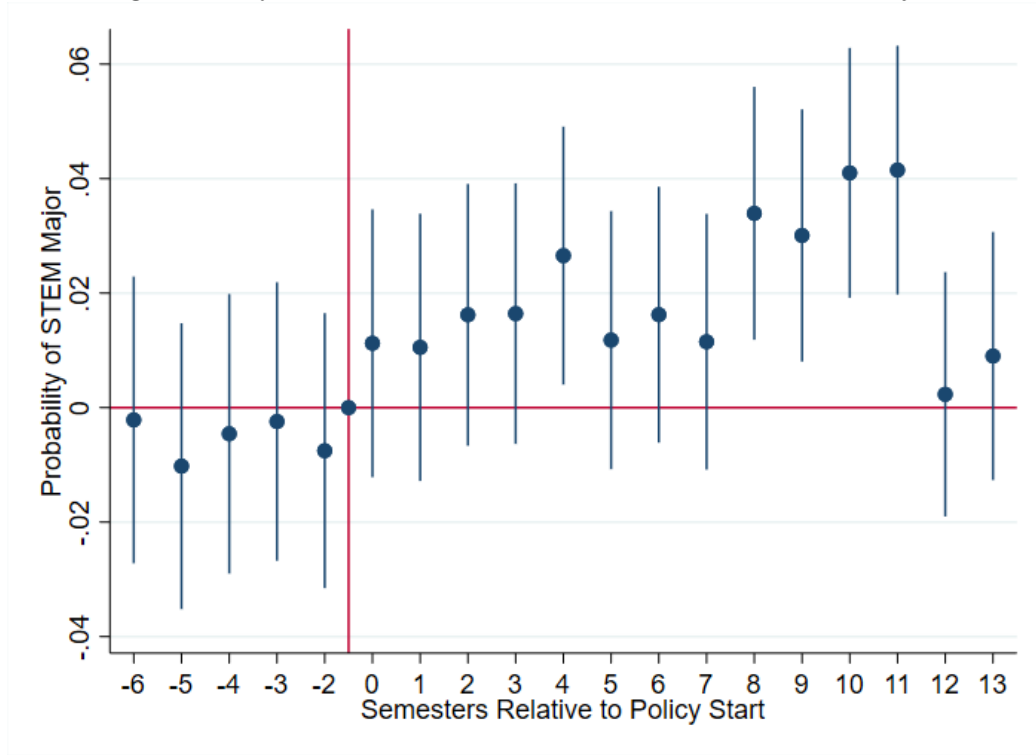
Notes: Sample includes all non-freshmen students enrolled from Fall 2004 through Spring 2014. The vertical dotted line indicates when the Math and Science Scholarship Enhancement program began to operate.

Figure 2: Average GPA Across STEM Majors



Notes: Average GPAs were calculated using data from Fall 2004 through Spring 2007 to capture proxied major difficulty prior to the enhancement program taking effect.

Figure 3: Dynamic Effects of Enhancements on STEM Majors



Notes: Event study estimates are generated using equation (1) described in the methodology section and replacing $Post_t$ with indicators for each semester before and after the enhancement program went into effect.

References

- Anderson, D. M., Broton, K. M. and Monaghan, D. B. (2023). Seeking STEM: The Causal Impact of Need-Based Grant Aid on Undergraduates' Field of Study, *Journal of Higher Education* .
- Budds, B. (2024). The teacher shortage solution South Carolina has been waiting for? New law offers financial boost. <https://www.wltx.com/article/news/education/new-law-offers-scholarships-address-sc-teacher-shortage/101-e0f69d4c-2c38-4d45-b2e6-d0322d0cbfe9>.
- Cornwell, C., Lee, K. H. and Mustard, D. B. (2006). The effects of state-sponsored merit scholarships on course selection and major choice in college, *Social Science Research Network* .
- Denning, J. T. and Turley, P. (2017). Was That SMART? Institutional Financial Incentives and Field of Study, *Journal of Human Resources* .
- Litchfield, S. (2024). About the South Carolina Education Lottery. <https://www.sceducationlottery.com/Lottery>.
- National Science Foundation (2024). Science and Engineering Indicators 2024: The State of U.S. Science and Engineering. <https://nces.nsf.gov/pubs/nsb20243>.
- Nietzel, M. T. (2023). States look to focus new funding on high demand college programs. <https://www.forbes.com/sites/michaelt Nietzel/2023/01/04/states-look-to-focus-new-funding-on-high-demand-college-programs/>.
- O'Rourke, B. (2021). Growing Gap in STEM Supply and Demand. <https://news.harvard.edu/gazette/story/2021/11/increasing-access-and-opportunity-in-stem-crucial-say-experts/>.
- Podobnik, B., Christopher Crawford, G., Lichtenstein, B., Lipic, T., Wild, D., Zhang, X. and Stanley, H. E. (2020). The new wealth of nations: How STEM fields generate the prosperity and inequality of individuals, companies, and countries, *Chaos, Solitons & Fractals* **141**: 110323. **URL:** <https://www.sciencedirect.com/science/article/pii/S0960077920307189>

- Sjoquist, D. L. and Winters, J. V. (2015). The effect of Georgia’s HOPE scholarship on college major: a focus on STEM, *IZA Journal of Labor Economics* .
- Smith, M. and Cortes, G. (2022). These are the highest- and lowest-paying STEM jobs in the U.S. <https://www.cnbc.com/2022/04/07/these-are-the-highest-and-lowest-paying-stem-jobs-in-the-us.html>.
- South Carolina Commission on Higher Education (2011). Ten-year summary of tuition fees for full-time, in-state undergraduates - public institutions, Spring 2011 update. <https://dc.statelibrary.sc.gov/entities/publication/7b829c8c-6285-4051-9392-3039118a201f>.
- U.S. Bureau of Labor Statistics (2024). Employment Projections: STEM Occupations. https://www.bls.gov/emp/tables/stem-employment.htm#BLStable_2024_4_11_10_25.
- Wang, X. (2013). Why Students Choose STEM Majors Motivation, High School Learning, and Postsecondary Context of Support, *American Educational Research Journal* .
- Williams, B. (2019). Who does aid help? examining heterogeneity in the effect of grant aid on achievement, <https://dx.doi.org/10.2139/ssrn.3216238> .